SAVEETHA SCHOOL OF ENGINEERING
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CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS **DATE: 22/07/2026**

EXPERIMENT 8 -- SELECT with various clause – GROUP BY, HAVING, ORDER BY

AIM:

To perform **Subqueries** and **Correlated Subqueries** on the **student_cgpa** table using MySQL Workbench.

PROCEDURE:

1. Open MySQL Workbench and connect to the MySQL server.
2. Select the required database using the USE command.
3. Create the student_cgpa table and insert the required records.
4. Display the table contents using the SELECT statement.
5. Execute **Subqueries** to retrieve records based on the result of another query.
6. Execute **Correlated Subqueries** that depend on values from the outer query.
7. Verify the output in the Result Grid.
8. Observe the results and compare them with the expected output.
9. Conclude the experiment after successful execution.

PROGRAM:

1. CREATE TABLE

```
CREATE TABLE student_cgpa (
reg_no VARCHAR(15) PRIMARY KEY,
student_name VARCHAR(50) NOT NULL,
department VARCHAR(20),
no_of_subs_pass INT,
no_of_subs_fail INT,
cgpa DECIMAL(3,2)
);
```

Then Insert Values:

```
INSERT INTO student_cgpa
(reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)
VALUES
('192411123','Deepthi','CSE',22,2,8.71),
('192512121','Varun','IT',27,1,9.23),
```

```
('192312117','Aadhya','ECE',25,3,8.97),
('192411227','Dimple','CSE',31,0,9.37),
('192820896','Mohan','EEE',18,9,6.48),
('192411122','Yogitha','CSE',31,1,9.01),
( '192516871','Sadhya','IT',25,5,7.22),
('192111101','Rahul','MECH',30,12,5.14),
('192411010','Alekhya','CSE',35,2,9.01),
('192511135','Rahul','CIVIL',37,0,9.72);

SELECT * FROM student_cgpa;
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192111101	Rahul	MECH	30	12	5.14
192312117	Aadhya	ECE	25	3	8.97
192411010	Alekhya	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411123	Deepthi	CSE	22	2	8.71
192411227	Dimple	CSE	31	0	9.37
192511135	Rahul	CIVIL	37	0	9.72
192512121	Varun	IT	27	1	9.23
192516871	Sadhya	IT	25	5	7.22
192820896	Mohan	EEE	18	9	6.48

#	Time	Action	Message	Duration / Fetch
1	22:41:47	CREATE TABLE student_cgpa (reg_no VARCHAR(15) PRIMARY KEY, student_name VARCHAR(50) NO...	0 row(s) affected	0.031 sec
2	22:44:31	INSERT INTO student_cgpa (reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa) VA...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	22:46:18	SELECT * FROM student_cgpa LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

1. Display students whose CGPA is greater than the average CGPA. (Subquery)

```
SELECT *
FROM student_cgpa
WHERE cgpa >
(
    SELECT AVG(cgpa)
    FROM student_cgpa
);
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192312117	Aadhya	ECE	25	3	8.97
192411010	Alekhya	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411123	Deepthi	CSE	22	2	8.71
192411227	Dimple	CSE	31	0	9.37
192511135	Rahul	CIVIL	37	0	9.72
192512121	Varun	IT	27	1	9.23
NULL	NULL	NULL	NULL	NULL	NULL

2. Display the student(s) having the highest CGPA. (Subquery)

```
SELECT *
FROM student_cgpa
WHERE cgpa =
(
   SELECT MAX(cgpa)
   FROM student_cgpa
);
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192511135	Rahul	CIVIL	37	0	9.72
NULL	NULL	NULL	NULL	NULL	NULL

3. Display students whose CGPA is greater than the average CGPA of their department. (Correlated Subquery)

```
SELECT *
FROM student_cgpa s1
WHERE cgpa >
(
   SELECT AVG(cgpa)
   FROM student_cgpa s2
   WHERE s1.department = s2.department
);
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192411227	Dimple	CSE	31	0	9.37
192512121	Varun	IT	27	1	9.23
NULL	NULL	NULL	NULL	NULL	NULL

4. Display students who have the highest CGPA in their department. (Correlated Subquery)

```
SELECT *
FROM student_cgpa s1
WHERE cgpa =
(
   SELECT MAX(cgpa)
```

```
    FROM student_cgpa s2
    WHERE s1.department = s2.department
);
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192111101	Rahul	MECH	30	12	5.14
192312117	Aadhya	ECE	25	3	8.97
192411227	Dimple	CSE	31	0	9.37
192511135	Rahul	CIVIL	37	0	9.72
192512121	Varun	IT	27	1	9.23
192820896	Mohan	EEE	18	9	6.48
NULL	NULL	NULL	NULL	NULL	NULL

RESULT

Thus, Subqueries and Correlated Subqueries were successfully performed on the student_cgpa table using MySQL Workbench.